Course notes

Meaning of a limit

$f(x) \to \ell$ when $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all x sufficiently large. For a limit at a, consider x of the domain, close to a and distinct from a. Left and right limits are written $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. When the domain has points on both sides of a, a two-sided limit exists if and only if the two one-sided limits exist and are equal. At an endpoint of the domain, we study the limit on the only accessible side.

Graphical interpretation

If $f(x) \to \ell \in \mathbb{R}$ when $x \to +\infty$ or $-\infty$, the line $y = \ell$ is a horizontal asymptote in this direction. If $f(x) \to \pm\infty$ when $x \to a$ on one side, the line $x = a$ is a vertical asymptote.

A curve can cross a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

References and operations

At $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. At $-\infty$, the sign of x^m depends on the parity of m. The rules for sum, product, quotient and comparison are those for sequences, with the same indeterminate forms.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. The limits at $\pm\infty$ are equal to 2. At 3^-, $f \to -\infty$; at 3^+, $f \to +\infty$. The asymptotes are $x = 3$ and $y = 2$.

Conjugate. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, be careful: $\sqrt{x^2} = |x| = -x$.

Exponential and comparative growth

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

We deduce $x^m e^{-x} \to 0$ at $+\infty$.

Proof of the limits of the exponential. The study of $e^x - x - 1$ gives $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ at $+\infty$. Setting $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Proof of a comparative growth result. For every integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If the inequality holds at order k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value at 0 is positive. The inductive step follows. By choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

Choose a transformation

Quotient of polynomials: factor out the highest powers. Difference of square roots: multiply by the conjugate. Quotient near a zero: factorise and study the signs. Exponential versus power: comparative growth, after a change of variable if necessary.

Comparison and composition: conditions of use

If $g(x) \leq f(x) \leq h(x)$ near the point or infinity studied, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (squeeze theorem). If $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit $-\infty$, we use an upper bound by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous at a, then $v(u(x)) \to v(a)$, when the composite is defined. For example $\sqrt{1+\frac{1}{x}} \to 1$ at $+\infty$. For an infinite limit of the outer function, check from which side $u(x)$ approaches the bound: $\frac{1}{u(x)}$ does not have the same behaviour according to the sign of $u(x)$.

Read a limit, choose a transformation

3.1 The approach side is part of the question

First use: S23.

Worked example · A vertical asymptote

For $f(x) = 1 + \frac{1}{x-2}$, the domain is $\mathbb{R} \setminus \{2\}$. When $x \to 2^-$, the denominator is negative and tends to 0, so $f(x) \to -\infty$. When $x \to 2^+$, $f(x) \to +\infty$. The line $x = 2$ is a vertical asymptote; no value $f(2)$ exists. At infinity, $f(x) \to 1$.

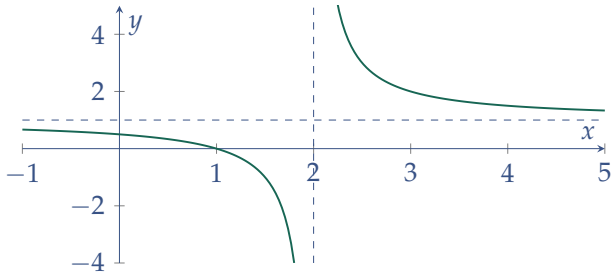


Figure 3.1. Find the two one-sided limits on the drawing: the branches are not joined.

A1 · Immediate application Determine the limits of $g(x) = 3 - \frac{2}{x+1}$ at -1^-, -1^+, $-\infty$ & $+\infty$. Give the asymptotes.

3.2 An indeterminate form calls for a calculation

First use: S24.

Worked example · The sign hidden in a square root

For $x < 0$, $\sqrt{x^2} = |x| = -x$. Thus, at $-\infty$,

$$\frac{\sqrt{x^2+1}}{x} = \frac{|x|\sqrt{1+1/x^2}}{x} = -\sqrt{1+1/x^2} \longrightarrow -1.$$

At $+\infty$, the same expression tends to 1. Omitting the absolute value would produce a positive limit for an expression that is actually negative.

A2 · Fading guidance Calculate at $+\infty$ the limits of $\frac{2x^2+1}{x^2-x}$, $\sqrt{x^2+4x} - x$ and $x\left(\sqrt{1+1/x} - 1\right)$. For the first, we suggest dividing by x^2 ; choose freely for the others.

A3 · Construct examples Construct three quotients whose limit has the form $0/0$ at 0, whose limits are respectively 0, 1 and $+\infty$. State the side of approach if necessary.

3.3 Comparative growth and change of variable

First use: S26; logarithm after S49.

Property · Exponential and powers · results from the lesson

For every integer $m \geq 1$, $e^x/x^m \to +\infty$ & $x^m e^{-x} \to 0$ at $+\infty$. To transfer these results, identify the new variable and its limit.

Worked example · Transferring a result

When $x \to -\infty$, set $t = -2x$ gives $t \to +\infty$ & $x^2e^{2x} = \frac{1}{4}t^2e^{-t} \to 0$. After S49, the substitution $t = 1/x$ in $x \ln x$ for $x \to 0^+$ gives $-(\ln t)/t \to 0$. The logarithm is not used before it has been studied.

A4 · Deferred application After S26, calculate $\lim_{x\to+\infty}(x^2+3x)e^{-x}$ & $\lim_{x\to-\infty} xe^x$. Say why formally replacing x by infinity is not a justification.

3.4 An asymptote may be crossed

First use: S24.

Definition · Horizontal Asymptote

If $f(x) - \ell \to 0$ at one of the infinities, $y = \ell$ is an asymptote at this infinity. This definition requires a vertical distance that tends to 0; it does not rule out any intersection at a finite distance.

Worked example · Two behaviours

On $\mathbb{R}$, $p(x) = 1 + 1/(1+x^2)$ remains strictly above $y = 1$. The function $q(x) = 1 + x/(1+x^2)$ crosses this same line at $x = 0$, while tending to 1 at both infinities. For q, the sign of $q(x) - 1$ is that of x.

A5 · Refute and explain A student says that the curve of q cannot cross its asymptote. Give the coordinates of the point of intersection and the signs on either side. Is this consistent with the two limits?

A6 · Choice of strategy Determine a, b so that $h(x) = \frac{ax+b}{x-1}$ has the horizontal asymptote $y = 2$ and satisfies $h(0) = 3$. Then study the one-sided limits at 1 and the position of the curve relative to $y = 2$.

One minute, spoken aloud. For each of the expressions in the "Methods" exercise, first state what prevents a conclusion from the operations on limits, then explain the role of the transformation.

Delayed review · S49 then S81. Redo the change of variable with $\ln x/x$; sort again the limits, mixing logarithms, exponentials and roots.

Investigation

A curve close to a line - 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ its curve in a coordinate system of the plane.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. Determine the limits of f at $-\infty$, at 1^-, at 1^+ and at $+\infty$.
3. Study the sign of $f(x) - 3$, then specify the position of $\mathcal{C}_f$ with respect to the line with equation $y = 3$.
4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0{,}01$.
5. Does curve $\mathcal{C}_f$ intersect its horizontal asymptote? Justify and explain why an overly wide graphing window can give a misleading impression.